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Turning Asthma Data Into Actionable Intelligence: A National Level Solution

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Asthma is a condition millions of people live with every day, and GPs collect loads of patient data on it to mitigate its impact. The information needed to spot deterioration early actually exists. The problem is no one is connecting it.



Imagine your school had three different offices and none of them talked to each other. Your principal doesn't know what your form tutor knows about you, and the

school counsellor also has a completely separate file on you. Now imagine that's how your health information works across your GP, your hospital, and your local clinic. That's what siloed data describes: Important information stuck in separate boxes that never get opened together.

Because doctors can only see part of the picture, they end up making decisions without all the facts. And the NHS ends up spending enormous amounts of money rushing to treat emergencies, when with better information sharing, many of those emergencies could have been spotted and stopped weeks earlier.

The Solution

Nobody has an asthma attack for just one reason. It's never just air pollution, or family history, or even just the time of year. It's a combination of these. It's like a set of dominoes that only fall when enough of them are lined up together. Most health systems only look at one domino at a time. The model we built made insightful predictions after looking at all of them at once.



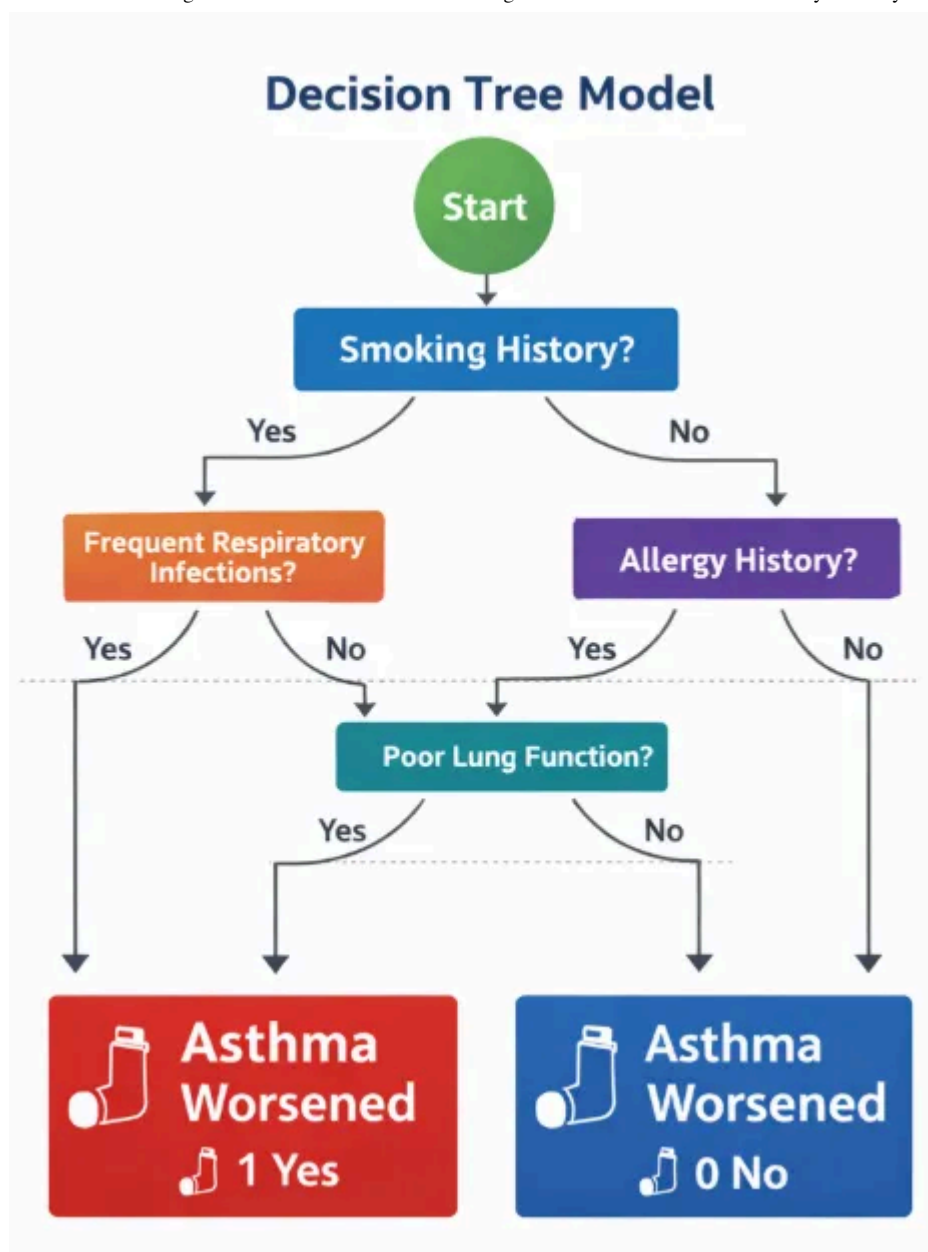
Think of it like this. Imagine you're trying to figure out why students keep getting sick at school. You could just look at their medical records, however that tells only a part of the story. What if you also looked at the school's proximity to a busy road or if it was hay fever season? Also, what if air quality in the area has been particularly

bad that week? Suddenly, you start to formulate a clearer picture. That's exactly what a model could do for asthma patients.

Selecting the correct model: Accuracy is not always the right answer



Selecting the right predictive model for this use case required a deliberate shift in how success was defined. In most data science contexts, accuracy is the primary metric; the higher its accuracy the better. In a clinical setting however, that instinct can be dangerous.



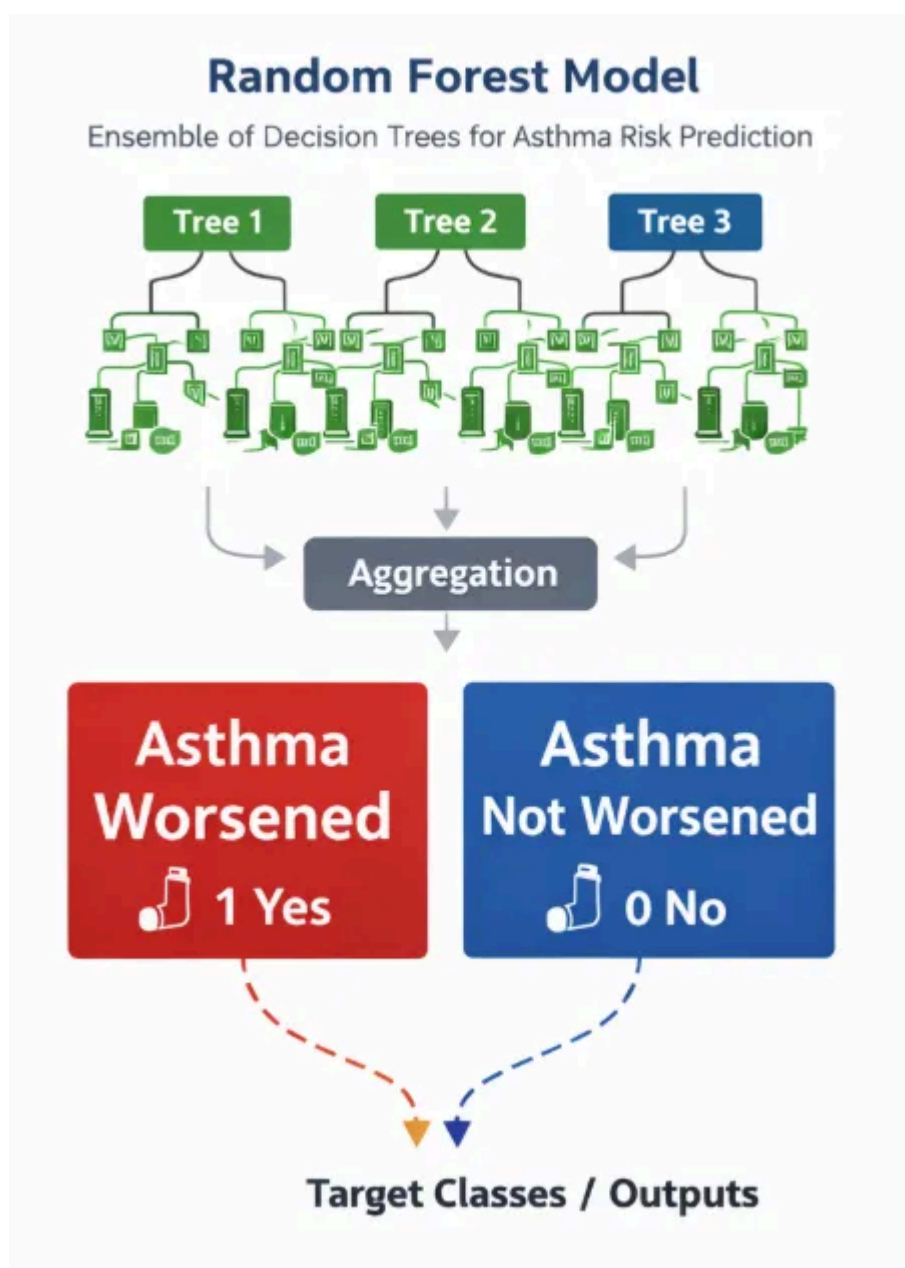
Decision Tree model to determine Asthma deterioration

The dataset reflected a real-world reality: the majority of asthma patients in any given population are not deteriorating at any given moment. That imbalance meant that a model could score an impressive 86% accuracy or higher, by simply predicting “no deterioration” for almost every patient, missing nearly every case that actually needed attention. Models like the Decision Tree and Logistic Regression fell into exactly that trap. Their accuracy looked really strong, but their ability to detect genuine deterioration was effectively zero. A tool that never raises the alarm is not a safety net, but a false one.

The evaluation, therefore, had to be centred on model sensitivity. This refers to the model’s ability to correctly identify patients who were genuinely worsening. In clinical terms, a missed deterioration case carries far greater consequences than a

false alarm. A false positive can be resolved with a follow-up appointment. A false negative can result in a preventable hospitalisation.

On that basis, the Random Forest model was the clear and necessary choice. Its sensitivity of 57.9% substantially outperformed the Decision Tree's near-zero detection rate and the Logistic Regression's complete failure to flag a single worsening case. The trade-off was a lower overall accuracy of 61.8%, which is a number that looks less impressive in isolation, but is far more clinically meaningful in context. The model was designed to catch the cases that mattered, not to inflate a headline figure.



The Random Forest Model created to predict Asthma deterioration

The outcome isn't a machine telling a doctor this person's 'asthma has worsened' only. Doctors already know that. Instead, it's more like an early warning system. You can allude it to a weather forecast, but instead of predicting rain, it's flagging which groups of patients are heading toward a danger zone before they even feel it themselves. That gives doctors and health teams the chance to intervene earlier, before things get serious.

Commercial Relevance: Where the Value Sits

The applications are immediate and scalable across several industry verticals. Insurance companies, for example, need to work out how likely their customers are to get seriously ill, because that affects how much things cost them. This model gives them a much more accurate way to do that. Instead of guessing, they can use real data to identify who needs support early, which means fewer expensive emergency claims down the line.

The Department of Health and Social Care at the national level who set the overall NHS budget for England, can use this to make smarter budget decisions. Instead of putting most of the money into treating emergencies after they happen, they can redirect it toward stopping those emergencies from happening in the first place. It's like fixing potholes before they destroy vehicle tyres, rather than paying for repairs afterwards.

Schools districts and their school health teams get something arguably the most tangible of all. By combining a student's health records with local air quality data, pollution levels near the school, and seasonal risk patterns, they get an early heads-up when a group of students is likely to struggle. That means support gets put in place before kids start missing classes and not after the damage to their attendance and grades is already done.

The Bottom Line

The data to predict asthma crises already exists. The analytical methods to extract signal from it are proven. What the industry now needs is the infrastructure to connect them, and the commercial clarity to recognise that prevention, at scale, is not just better medicine, but rather sound economics.



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